

# Optimal control of a fed-batch fermentation process

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*Optimisation of fed-batch fermentation processes usually employs the calculus of variations to determine optimal feed-rate profiles that will maximise a given objective function. This results in a two-point boundary-value problem and because of the nonlinear nature of the processes, the optimal solution usually falls out as an open-loop control algorithm. One advantage of this approach is that it does not need measurements of state variables which are often difficult to obtain on-line. Instead it assumes that the state variables are proceeding along known paths a-priori determined by models. However, the disadvantage of such an approach is that the performance will severely deteriorate in the presence of process disturbances or plant-model mismatch. Recently, on-line estimation of state variables has been successfully developed and used in the industry and therefore a method which operates the fed-batch fermentation in a closed-loop control scheme using state feedback is proposed. This is achieved by a two-step method. First, the optimal substrate concentration profile which governs the biochemical reactions in the fermentation process is determined. Then a controller is designed in closed-loop form to track this desired profile. Simulation studies for both primary and secondary metabolite production processes show that better performance is obtained by this closed-loop optimal control method than by the open-loop optimal feed rate profile method. This is due to the self-correcting property of the proposed method, which proves to be advantageous when there are disturbances or plant-model mismatch.*

**Keywords:** Fed-batch fermentation; optimal control; process optimisation; calculus of variations; primary metabolite; secondary metabolite.

## Notation

|                  |                                                           |
|------------------|-----------------------------------------------------------|
| $F$              | Substrate feed rate (l/hr)                                |
| $F_{\max}$       | Maximum substrate feed rate (l/hr)                        |
| $H$              | Hamiltonian                                               |
| $S$              | Substrate concentration (g/l)                             |
| $S_{\text{opt}}$ | Optimal substrate concentration (g/l)                     |
| $S_f$            | Substrate concentration in the feed stream (g/l)          |
| $X$              | Biomass concentration (g/l)                               |
| $P$              | Product concentration (mg/l)                              |
| $V$              | Culture volume (l)                                        |
| $V_f$            | Maximum culture volume (l)                                |
| $D$              | Dilution rate ( $\text{hr}^{-1}$ )                        |
| $Y_{\text{xs}}$  | Yield of biomass from substrate (g. biomass/g. substrate) |

|                            |                                                              |
|----------------------------|--------------------------------------------------------------|
| $u$                        | Control variable                                             |
| $t$                        | Time (hr)                                                    |
| $t_f$                      | Final time                                                   |
| $\mu$                      | Specific growth rate ( $\text{hr}^{-1}$ )                    |
| $\mu_{\max}$               | Maximum specific growth rate ( $\text{hr}^{-1}$ )            |
| $\pi$                      | Specific product formation rate ( $\text{hr}^{-1}$ )         |
| $\pi_{\max}$               | Maximum specific product formation rate ( $\text{hr}^{-1}$ ) |
| $K_s, K_i, K_{ms}, K_{pi}$ | Kinetic constants (g/l)                                      |
| $\lambda_i$                | Costate variable, $i = X, S, P, V$                           |

## 1. Introduction

Fermentation processes are used for producing many fine chemical substances such as amino acids, antibiotics, baker's yeast, enzymes, etc. Amongst the modes of operation (batch, fed-batch and continuous), the fed-batch technique is often used in industry due to its ability to overcome the catabolite repression or glucose effect, which usually occurs during production of these fine chemicals (Parulekar and Lim, 1985; Yamane and Shimizu, 1984). Fed-batch operation also gives the operator the degree of freedom to manipulate the process via substrate feed rates. Thus arises the challenge to control and optimise the fed-batch fermentation processes.

Optimisation of fed-batch fermentation processes has been a topic of research for many years. From a biological side, a strain improvement of micro-organism and media design (Huber and Tietz, 1983; Kennedy *et al.*, 1992) are usually employed to optimise the fermentation process. This also includes the use of optimal experimental design such as the Evolutionary Operation (EVOP) method (Banerjee, 1993) as well as a new design of a bioreactor. However, from a process operational point of view to enhance the process optimisation further, the most used approach for process optimisation is to calculate an optimal feed-rate profile that will optimise a given objective function. To determine the optimal feed-rate profile in fed-batch fermentation, the other environmental variables such as temperature and pH, which also affect the bioreaction rates in the processes, are assumed constant at some levels. The approaches used by many research groups to determine the substrate feed-rate profile that optimises a desired objective function are usually based on the calculus of variations (Weigand *et al.*, 1979; Cazzador, 1988; Modak *et al.*, 1986; Lim *et al.*, 1986; Takamatsu *et al.*, 1995; Shimizu *et al.*, 1991) or Greens function (Ohno *et al.*, 1976). A recent survey on the process optimisation can be found in Terwiesch *et al.* (1994).

Since there are physical constraints on the minimum and maximum feed rates, the Pontryagin's Maximum

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principle is also applied. However, there are two problems arising in applying the variational method to the fermentation process. The first is that a singular control situation may occur during operation since the control input or feed rate may appear explicitly linear in the Hamiltonian. Hence, the Maximum principle fails to provide a solution. The other problem is that the obtained optimal feed rate profile from the variational method is implemented in an open-loop control structure and can suffer quite severely when the model parameters are not exactly known, as in real applications. This method, nevertheless, requires no measurement of state variables in the process.

Since many state variables in the fermentation processes such as biomass, substrate and product concentration are difficult to measure on-line, many methods have therefore been developed for on-line estimation of these state variables. These state estimation methods are usually based on linear or nonlinear Kalman filters or neural networks (Stephanopoulos and San, 1982; Park *et al.*, 1983; Bastin and Dochain, 1986; Shimizu *et al.*, 1986; Zeng *et al.*, 1991; Bellgardt *et al.*, 1986; Nahlik and Burianec, 1988; Liu *et al.*, 1992).

As the on-line neural network estimation of substrate concentration has recently been successfully developed and implemented in industry (Zhang *et al.*, 1996), it is now possible to develop a method which can overcome the described shortcomings in the conventional optimisation method. The proposed method separates the optimisation problem of a fermentation process into two parts. Firstly, the optimal substrate concentration profile which has direct effect on the biochemical reaction rates in the fermentation process is derived. Then, a controller is designed to track the obtained optimal substrate concentration profile. With this approach, the singular problem is avoided since the substrate concentration appears as a nonlinear function in the Hamiltonian. The open-loop control is also transformed into a closed-loop servo or regulator control. Since the process is operated in closed loop, this method is here called 'closed-loop optimal control'.

In this paper, the two-step optimisation method is presented. The similarities and differences as well as advantages and disadvantages between the proposed method and the conventional one are also discussed. The rest of the paper is structured as follows: Section 2 describes the mathematical representation of a fed-batch fermentation process. The formulation of the closed-loop optimal control is illustrated in Section 3. This is followed by simulations of primary and secondary metabolite production in Section 4. The discussion is in Section 5 and the paper is concluded in Section 6.

## 2. Mathematical representation of a fed-batch fermentation process

Mathematical models of a fed-batch fermentation process can be written based on mass balance equations as:

$$\frac{dX}{dt} = \mu X - DX \quad \dots (1)$$

$$\frac{dS}{dt} = -\frac{1}{Y_{xs}} \mu X + D(S_f - S) \quad \dots (2)$$

$$\frac{dP}{dt} = \pi X - DP \quad \dots (3)$$

$$\frac{dV}{dt} = F \quad \dots (4)$$

$$D = F/V \quad \dots (5)$$

where  $X$ ,  $S$ ,  $P$  are biomass, substrate and product concentration in the reactor respectively;  $F$  is substrate feed rate;  $S_f$  is the concentration of substrate in the feed stream;  $D$  is dilution rate;  $\mu$  and  $\pi$  are specific cell growth rate and specific product formation rate, respectively;  $Y_{xs}$  is yield of cell mass from substrate and  $V$  is culture volume. The specific rates  $\mu$  and  $\pi$  are usually functions of substrate concentration. Further details and analysis on fed-batch operation can be found in Yamane and Shimizu (1984), Dunn and Mor (1975) and Lim *et al.* (1977). Eqns (1)–(5) represent a general model for a secondary metabolite production. For a primary metabolite process, since the primary metabolite depends directly on the biomass, and biomass can also be referred to as one of the primary metabolite, Eqn (3) can then be omitted. Eqns (1), (2), (4) and (5) then constitute a model for the primary metabolite production in which biomass is the product.

The fed-batch fermentation process is constrained by conditions on the maximum culture volume and minimum and maximum values of the substrate feed rate:

$$0 \leq F \leq F_{\max} \quad \dots (6)$$

$$V(t_f) = V_f \quad \dots (7)$$

In the literature, the objective function is usually given as:

$$J(F) = f(X(t_f), P(t_f)) \quad \dots (8)$$

in which the substrate feed rate is chosen to maximise or minimise a function of final biomass or product concentration (eg. maximise the biomass in a Baker yeast fermentation or maximise a secondary metabolite in an antibiotic fermentation).

The optimal feed rate is usually obtained as a time sequence due to the fact that the process model is nonlinear. Therefore, the optimal control strategy is usually implemented in open loop.

## 3. Closed-loop optimal control

Here, a method in which the optimal control can be implemented in closed loop is proposed. To achieve this, the proposed method separates an optimal control problem in a fed-batch fermentation process into two parts. The first part is to determine the optimal environmental variable. In this paper, this role is taken by the substrate concentration in the fermenter. The optimal substrate concentration profile is determined by the calculus of variations method. This avoids the singular problem which usually appears during the operation as the Hamiltonian is a nonlinear function of the substrate concentration which is used as a control variable in this method. In the second part, a controller is designed to track the obtained optimal substrate profile. The control action is thus implemented in closed loop. This offers the flexibility that many types of

controllers can be designed and used, particularly those robust to model/process mismatch errors (Green and Limebeer, 1994). As the substrate feed rate is used directly to control the substrate concentration level in the fermenter, it is more transparent than using the substrate feed rate to control the product concentration.

These two parts are complementary with each other. In designing the optimal substrate profile, both substrate feed rate and volume are omitted from the system models. The controller is then not only used to cope with the omitting of feed rate and volume constraints but also to track the desired substrate concentration profile, which in turn optimises the given objective function.

### 3.1 Optimal substrate profile determination

In this subsection, the optimal substrate concentration, which is used as a control variable to optimise the objective function in the proposed method, is determined. As mentioned earlier, although the substrate feed rate and volume have been omitted here, they will be incorporated in the controller design part in the next subsection. The system equations then reduce to:

$$\frac{dX}{dt} = \mu X \quad \dots (9)$$

$$\frac{dP}{dt} = \pi X \quad \dots (10)$$

The process objective is to optimise a function of biomass or product concentration at the final operating time; eg, maximising the product concentration at the final time. The objective function is then written as:

$$J(S) = f(X(t_f), P(t_f)) \quad \dots (11)$$

in which the profile of substrate concentration is chosen to minimise or maximise the biomass or the desired product at the final operating time.

The calculus of variations is used to determine the optimal substrate concentration profile. In this case, the Hamiltonian and costate equations can be written as:

$$H = \lambda_X \mu X + \lambda_P \pi X \quad \dots (12)$$

$$\frac{d\lambda_X}{dt} = -\frac{\partial H}{\partial X} = \lambda_X \mu + \lambda_P \pi \quad \dots (13)$$

$$\frac{d\lambda_P}{dt} = -\frac{\partial H}{\partial P} = 0 \quad \dots (14)$$

Transversality conditions:

$$\lambda_X(t_f) = \frac{\partial J}{\partial X(t_f)} \quad \dots (15)$$

$$\lambda_P(t_f) = \frac{\partial J}{\partial P(t_f)} \quad \dots (16)$$

The optimal substrate concentration profile satisfies the following equation:

$$\frac{\partial H}{\partial S} = \lambda_X X \frac{\partial \mu}{\partial S} + \lambda_P X \frac{\partial \pi}{\partial S} = 0 \quad \dots (17)$$

### 3.2 Controller design

Once the optimal substrate concentration profile is determined, a controller is designed to track this profile by manipulating the substrate feed rate. There is no restriction on the types of controllers being used as long as they can track the desired profile or set-point well. This is another advantage of this method for the flexibility in choosing between many types of controllers.

A nonlinear model predictive control scheme for the tracking control component is used in the proposed method. Model predictive control is defined (Biegler, 1991; Sistu and Bequette, 1991) as a control scheme in which the controller determines a control variable profile that optimises some open-loop performance objective on a time interval extending from the current time to the current time plus a prediction horizon. Feedback is incorporated when a further measurement becomes available and the optimisation procedure for determining the control variable profile is then restarted. This characteristic is called a receding horizon or finite moving horizon approach. The general problem to be solved by model predictive control can be stated as:

$$\min_{u(t)} \Phi[u(t), x(t), y(t)]$$

subject to the system equations, constraints and initial conditions:

$$\frac{dx}{dt} - f(x, u) = 0$$

$$y - g(x, u) = 0$$

$$h(x, u) = 0$$

$$k(x, u) \geq 0$$

$$x(t_0) = x_0$$

with:

$$t \in [t_0, t_0 + T]$$

The notation here refers only to this subsection:  $u$  is the control input,  $y$  is the output and  $x$  is the state. The time interval is from the current time  $t_0$  to  $t_0 + T$ , in which  $T$  is the length of the prediction horizon. The functional  $\Phi$  is the performance objective of the controller. The functions  $f$  and  $g$  represent the process models, and  $h$  and  $k$  are equality and inequality constraints. Model predictive control is therefore capable of handling the process constraint explicitly. Further details on nonlinear model predictive control are given in Biegler (1991), Sistu and Bequette (1991), Ali and Zafiriou (1993), Cuthrell and Biegler (1987), Renfro *et al* (1987), and Eaton and Rawlings (1990).

The nonlinear model predictive control applied to a fed-batch fermentation process under the proposed control scheme can then be stated as follows where the substrate feed rate ( $F$ ), which is the control variable, can be obtained:

$$\min_{F(k) \dots F(k+m-1)} \sum_{i=1}^p (\hat{S}(k+i/k) - S_{opt}(k+i))^2$$



where

$\hat{S}(k+i/k)$ : model predicted value of substrate concentration at time  $(k+i)$  based on information at time  $k$ .

$S_{\text{opt}}(k+i)$ : optimal substrate concentration at time  $(k+i)$ .

$p$ : prediction horizon

$m$ : control horizon:  $(F(k+i) = 0 \quad \forall i \geq m; m < p)$

subject to:

$$\frac{dX}{dt} = \mu X - DX; \quad X(0) = X_0$$

$$\frac{dS}{dt} = -\frac{1}{Y_{xs}} \mu X + D(S_f - S); \quad S(0) = S_0$$

$$\frac{dP}{dt} = \pi X - DP; \quad P(0) = P_0$$

$$\frac{dV}{dt} = F; \quad V(0) = V_0$$

$$0 \leq F \leq F_{\text{max}}$$

$$V(t_f) \leq V_f$$

with:

$$t \in [t_0, t_0 + T]$$

#### 4. Optimal control of primary and secondary metabolite production processes

The proposed closed-loop optimal control method is illustrated with application to primary and secondary metabolite production in this section.

In the primary metabolite production process, primary metabolites are synthesised directly from the primary metabolism. Examples of the primary metabolites are protein, ethanol, gluconic acid and other cell constituents. Biomass production is therefore referred to as one of primary metabolite production.

In the secondary metabolite production process, secondary metabolite production is associated with limited or sub-optimal growth. There are generally two phases taking place for the secondary metabolite production: growth phase and production phase. The process starts with microbial growth and primary metabolism, and secondary products are formed afterwards. The production of secondary metabolites is not necessarily started after the growth has stopped or entirely separated from the growth phase. Many antibiotics and vitamins belong to the secondary metabolites.

The simulations for both metabolite production are based on the fermentation models described earlier. (Eqns (1)–(5) are used as the secondary metabolite production model and Eqns (1), (2), (4) and (5) are used as the primary metabolite production model.) The specific growth rate ( $\mu$ ) and the specific product formation rate ( $\pi$ ) in the models are functions of substrate concentration and in the form of substrate inhibition

kinetic (Yamane and Shimizu, 1984; Han and Levenspiel, 1988; Hong, 1986), which are shown in Eqns (18) and (19):

$$\mu = \frac{\mu_{\text{max}} S}{\left(K_s + S + \frac{S^2}{K_i}\right)} \quad \dots (18)$$

$$\pi = \frac{\pi_{\text{max}} S}{K_{\pi s} + S + \frac{S^2}{K_{\pi i}}} \quad \dots (19)$$

The substrate inhibition kinetic is employed here because it can represent the catabolite repression effect, which in turn requires the operation of fermentation in the fed-batch mode. The parameters of the models are tabulated in Table 1 for the primary metabolite process and in Table 2 for the secondary metabolite process. Note that the parameters used for simulation are chosen arbitrarily without any intention to be specific for a particular process but for representing a general characteristic for a class of processes with the substrate inhibition type kinetic.

TABLE 1: Parameters used in the simulation for a primary metabolite production

| Parameter          | Value | Unit                         |
|--------------------|-------|------------------------------|
| $\mu_{\text{max}}$ | 0.10  | (g biomass/(g biomass * hr)) |
| $K_s$              | 3.0   | (g substrate/litre)          |
| $K_i$              | 8.34  | (g substrate/litre)          |
| $Y_{xs}$           | 0.164 | (g biomass/g substrate)      |
| $X(0)$             | 1     | (g biomass)                  |
| $S(0)$             | 20    | (g substrate)                |
| $V(0)$             | 20    | (litre)                      |
| $V(\text{tf})$     | 50    | (litre)                      |
| $S_f$              | 100   | (g substrate/litre)          |

TABLE 2: Parameters used in the simulation for a secondary metabolite production

| Parameter          | Value | Unit                          |
|--------------------|-------|-------------------------------|
| $\mu_{\text{max}}$ | 0.10  | (g biomass/(g biomass * hr))  |
| $K_s$              | 3.0   | (g substrate/litre)           |
| $K_i$              | 8.34  | (g substrate/litre)           |
| $Y_{xs}$           | 0.164 | (g biomass/g substrate)       |
| $\pi_{\text{max}}$ | 0.25  | (mg product/(g biomass * hr)) |
| $K_{\pi s}$        | 0.4   | (g substrate/litre)           |
| $K_{\pi i}$        | 10    | (g substrate/litre)           |
| $X(0)$             | 1.0   | (g biomass)                   |
| $S(0)$             | 4.6   | (g substrate)                 |
| $V(0)$             | 20    | (litre)                       |
| $V(\text{tf})$     | 50    | (litre)                       |
| $S_f$              | 100   | (g substrate/litre)           |

#### 4.1 Optimal substrate profile determination

The optimal substrate concentration profile is determined by applying the calculus of variations method as shown in the previous section. The optimal substrate concentration is obtained by solving Eqn (17).

#### 4.1.1 Primary metabolite production

A biomass production is employed here as an example of a primary metabolite production process. The objective function here is to maximise the biomass concentration at the final time. The process model and the objective function are written as:

$$\frac{dX}{dt} = \mu X \quad \dots (20)$$

$$J(S) = X(t_f) - \varepsilon \int_{t_0}^{t_f} dt \quad \dots (21)$$

where  $\varepsilon$  is the cost factor per unit of operating time. The Hamiltonian and costate equations are then written as:

$$H = -\varepsilon + \lambda_X \mu X \quad \dots (22)$$

$$\frac{d\lambda_X}{dt} = -\frac{\partial H}{\partial X} = -\lambda_X \mu \quad \dots (23)$$

with the transversality condition:

$$\lambda_X(t_f) = \frac{\partial J}{\partial X(t_f)} = 1 \quad \dots (24)$$

The optimal substrate concentration profile is calculated from:

$$\frac{\partial H}{\partial S} = \lambda_X X \frac{\partial \mu}{\partial S} = 0 \quad \dots (25)$$

which implies that:

$$\frac{\partial \mu}{\partial S} = 0 \quad \dots (26)$$

The optimal substrate concentration ( $S_{opt}$ ), which satisfies Eqn (26) is then obtained as:

$$S_{opt} = \sqrt{K_S \cdot K_i} = 5 \text{ g/l} \quad \dots (27)$$

The optimal substrate concentration profile in this case is to be kept constant at the level which maximises the specific growth rate (Eqn (26)). This can be explained by the fact that for maximum productivity of the primary metabolite production, the substrate concentration should be kept at the level such that the specific growth rate is maximal.

#### 4.1.2 Secondary metabolite production

For a secondary metabolite production, the objective function is to maximise the final product concentration. The process model and the objective function are written as:

$$\frac{dX}{dt} = \mu X \quad \dots (28)$$

$$\frac{dP}{dt} = \pi X \quad \dots (29)$$

$$J(S) = P(t_f) - \varepsilon \int_{t_0}^{t_f} dt \quad \dots (30)$$

where  $\varepsilon$  is the cost factor per unit of operating time. The Hamiltonian and costate equations can then be written as:

$$H = -\varepsilon + \lambda_X \mu X + \lambda_P \pi X \quad \dots (31)$$

$$\frac{d\lambda_X}{dt} = -\frac{\partial H}{\partial X} = -\lambda_X \mu - \lambda_P \pi \quad \dots (32)$$

$$\frac{d\lambda_P}{dt} = -\frac{\partial H}{\partial P} = 0 \quad \dots (33)$$

Transversality conditions:

$$\lambda_X(t_f) = \frac{\partial J}{\partial X(t_f)} = 0 \quad \dots (34)$$

$$\lambda_P(t_f) = \frac{\partial J}{\partial P(t_f)} = 1 \quad \dots (35)$$

The optimal substrate concentration for the secondary metabolite production can be calculated from:

$$\frac{\partial H}{\partial S} = \lambda_X X \frac{\partial \mu}{\partial S} + \lambda_P X \frac{\partial \pi}{\partial S} = 0 \quad \dots (36)$$

which implies that:

$$\lambda_X \frac{\partial \mu}{\partial S} + \lambda_P \frac{\partial \pi}{\partial S} = 0 \quad \dots (37)$$

Since the Hamiltonian ( $H$ ) is zero for a free-time problem (final operating time is not fixed), the Hamiltonian equation (31) becomes:

$$H = -\varepsilon + \lambda_X \mu X + \lambda_P \pi X = 0 \quad \dots (38)$$

or

$$X = \frac{\varepsilon}{\lambda_X \mu + \lambda_P \pi} \quad \dots (39)$$

Finding the value of  $\lambda_X$  in Eqn (37) and substituting into Eqn (39) yields:

$$X = \frac{\mu' \varepsilon}{-\pi' \mu + \pi \mu'} \quad \dots (40)$$

where ' denotes the differentiation with respect to the substrate concentration ( $S$ ) and  $\mu'$  and  $\pi'$  can be expressed as:

$$\frac{d\mu}{dS} = \frac{\mu_{\max}}{K_i} (K_S K_i - S^2) \quad \dots (41)$$

$$\frac{d\pi}{dS} = \frac{\pi_{\max}}{K_{\pi i}} (K_{\pi s} K_{\pi i} - S^2) \quad \dots (42)$$

Eqn (40) therefore shows clearly a relationship between the biomass and substrate concentration at different cost factors ( $\varepsilon$ ). The relation between the biomass and substrate concentration at different cost factors ( $\varepsilon$ ) is shown in Fig 1. The far left line shows the smaller cost factor, while the far right shows the bigger cost factor.

Taking the time derivative of Eqn (37) yields:

$$-\lambda_X \mu \mu' - \lambda_P \pi \mu' + \lambda_X \mu'' \dot{S} + \lambda_P \pi'' \dot{S} = 0 \quad \dots (43)$$

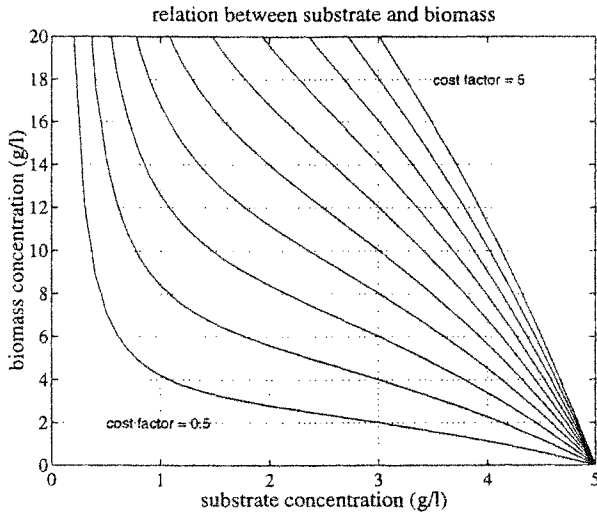


Fig 1 Relationship between biomass and substrate concentration at different cost factor ( $\epsilon$ ). The far left line shows the smaller cost factor, while that on the far right shows the bigger cost factor

Substituting with  $\lambda_x$  from (37) and  $\lambda_p$  from (33) and (35) results in:

$$\dot{S} = -\frac{\mu'(\mu\pi' - \pi\mu')}{(\mu'\pi'' - \pi'\mu'')} \quad \dots (44)$$

The optimal substrate concentration profile described by Eqn (44) is shown in Fig 2. There are two special cases here. To make the substrate concentration in Eqn (44) constant, the following two conditions exist:

$$\mu' = 0 \quad \dots (45)$$

or

$$(\mu\pi' - \pi\mu') = 0 \quad \dots (46)$$

From Fig 2, the optimal substrate concentration starts from the condition suitable for the microbial growth ( $\mu' = 0$ ) to the condition suitable for product formation ( $d(\pi/\mu)/dS = 0$ ) (see also discussion in Section 5.2). The optimal substrate concentration profile in this case varies within these two values and therefore is bounded with the following condition:

$$S_{(d(\pi/\mu)/dS=0)} < S < S_{(\mu'=0)}$$

## 4.2 Controller design

The nonlinear model predictive control is used in the proposed closed loop optimal control method with the following objective function as shown in Section 3.2:

$$\min_{F(k) \dots F(k+m-1)} \sum_{i=1}^p (\hat{S}(k+i/k) - S_{opt}(k+i))^2$$

where:

$\hat{S}(k+i/k)$ : model predicted value of substrate concentration at time  $(k+i)$  based on information at time  $k$ .

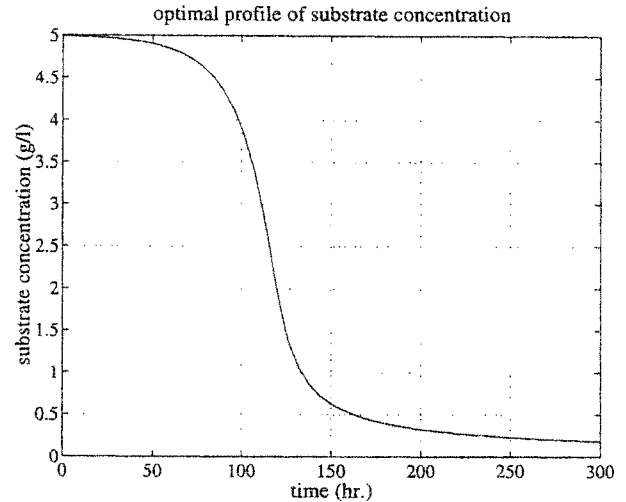


Fig 2 Relationship between optimal substrate concentration and time for the secondary metabolite. The optimal substrate concentration ranges between that suitable for the microbial growth ( $\mu' = 0$ ) and that suitable for product formation ( $d(\pi/\mu)/dS = 0$ )

$S_{opt}(k+i)$ : optimal substrate concentration at time  $(k+i)$ .  
 $p$ : prediction horizon.  
 $m$ : control horizon: ( $F(k+i) = 0 \quad \forall i \geq m$ ;  $m < p$ ).

The following design parameters are used in the controller: sampling time, 1 hr; prediction horizon ( $p$ ), 5 hr; control horizon ( $m$ ), 3 hr. These design parameters are chosen such that the controller can perform sufficiently well without excessive computation burden. (The sampling time of 1 hr is small enough for a slow dynamical system such as the fermentation process while the prediction horizon and control horizon of 5 and 3 hr are also long enough to control the process.)

All simulations (cost factor ( $\epsilon$ ) = 1) are performed in the Matlab and Simulink environment. Determination of the optimal control signal at each time interval employs the successive quadratic programming method, which is available in the constrained optimisation routine in Matlab.

The profiles of variables under the nonlinear model predictive control in the primary and secondary metabolite production are shown in Figs 3 and 4, respectively. Note that these simulations are performed assuming that the perfect models for both processes are known and therefore resulting in the optimal production of the metabolites.

## 5. Methods comparison and discussion

As the closed-loop optimal control of a fed-batch fermentation process is developed in this paper, the performances of the proposed method and the conventional method are compared. The effect of cost factor per unit of operating time is also discussed.

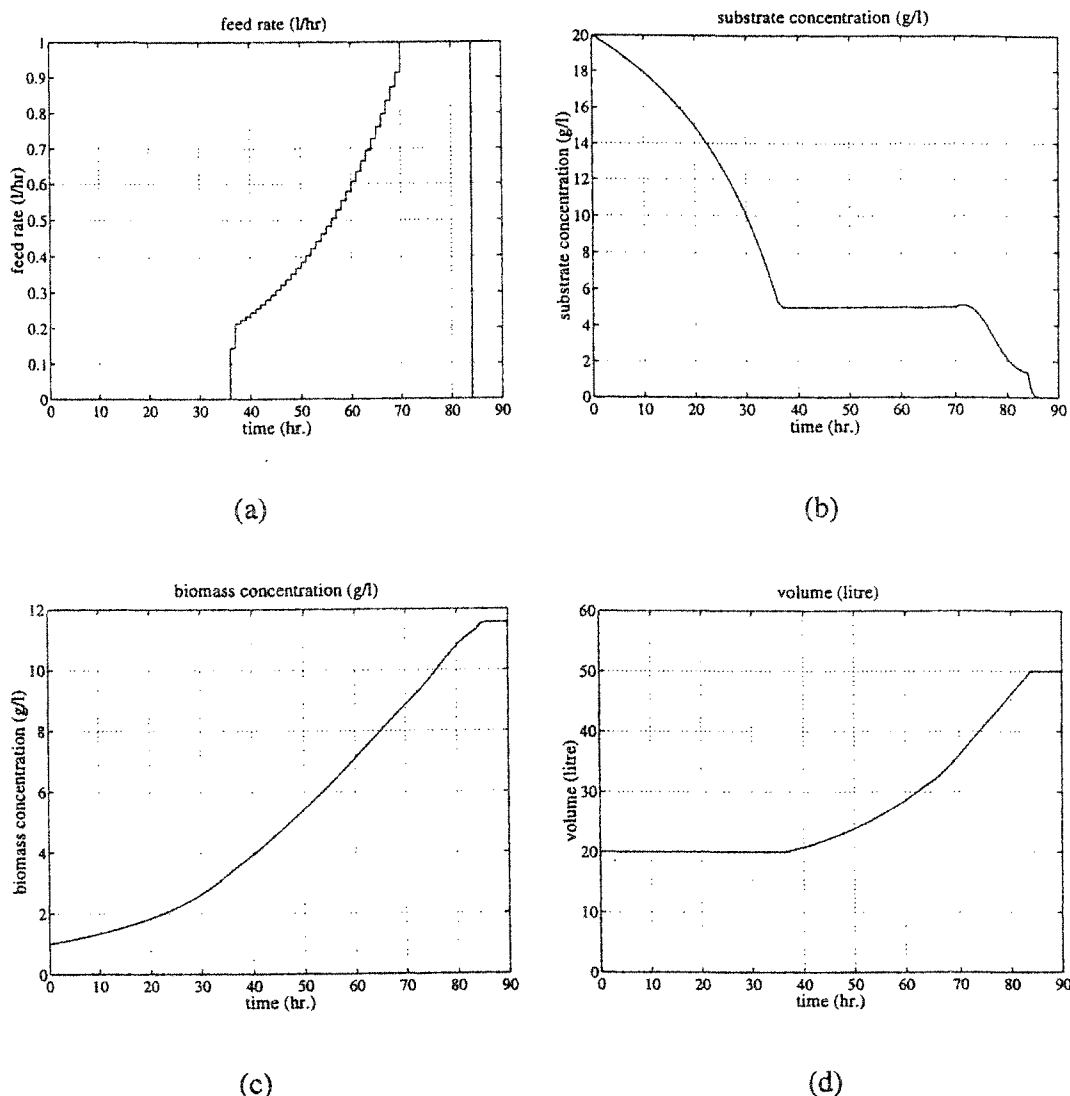


Fig 3 Simulation of closed-loop optimal control for a primary metabolite process (assuming perfect model). In the absence of plant-model mismatch or disturbance, both open-loop optimal feed rate and closed-loop optimal control algorithms lead to identical results

### 5.1 Comparison between optimal substrate concentration profile and optimal feed rate profile method

Details on a conventional fed-batch optimisation method in which an open-loop optimal feed rate profile is determined can be found in Modak *et al* (1986) and Lim *et al* (1986). Here, the proposed closed-loop optimal control method is compared with the open-loop optimal feed rate profile method. Block diagrams comparing the structure of both methods are shown in Figs 5 and 6.

For comparison purposes, the simulations are performed for a plant-model mismatch case. In a perfect model case, similar variable profiles are obtained from both methods for primary and secondary metabolite processes. It is not surprising in this case as the controller in the closed-loop method regulates the substrate concentration to follow the optimal profile. While, for the open-loop method, the optimal feed rate sequence also corresponds to maintaining the substrate concentration at the optimal level. With the exact model

parameters, both methods give similar profiles of variables. However, the closed-loop optimal control method is more transparent and flexible as any controllers can be used so long as they can achieve the objective of keeping substrate concentration at a given optimum level or profile to maximise primary (biomass) or secondary metabolite in fed-batch fermentation processes. In the open-loop optimal feed rate profile method, feed rate sequences are pre-determined to maximise the given objective function in one step. Therefore, the optimal substrate concentration level is not explicitly shown and hence is opaque to the operator.

In the case of plant-model mismatch, better performance for the proposed method than that for the open-loop method is shown in Figs 7 and 8 for the simulations of the primary and secondary metabolite processes. In the simulations, the parameter  $Y_{xs}$  used in the model is 0.149 while it equals 0.164 in the plant. For the primary metabolite process, the miscalculation of the feed rate profile due to the incorrect parameter results in the process being prolonged to 85 hr (comparing to 75 hr by

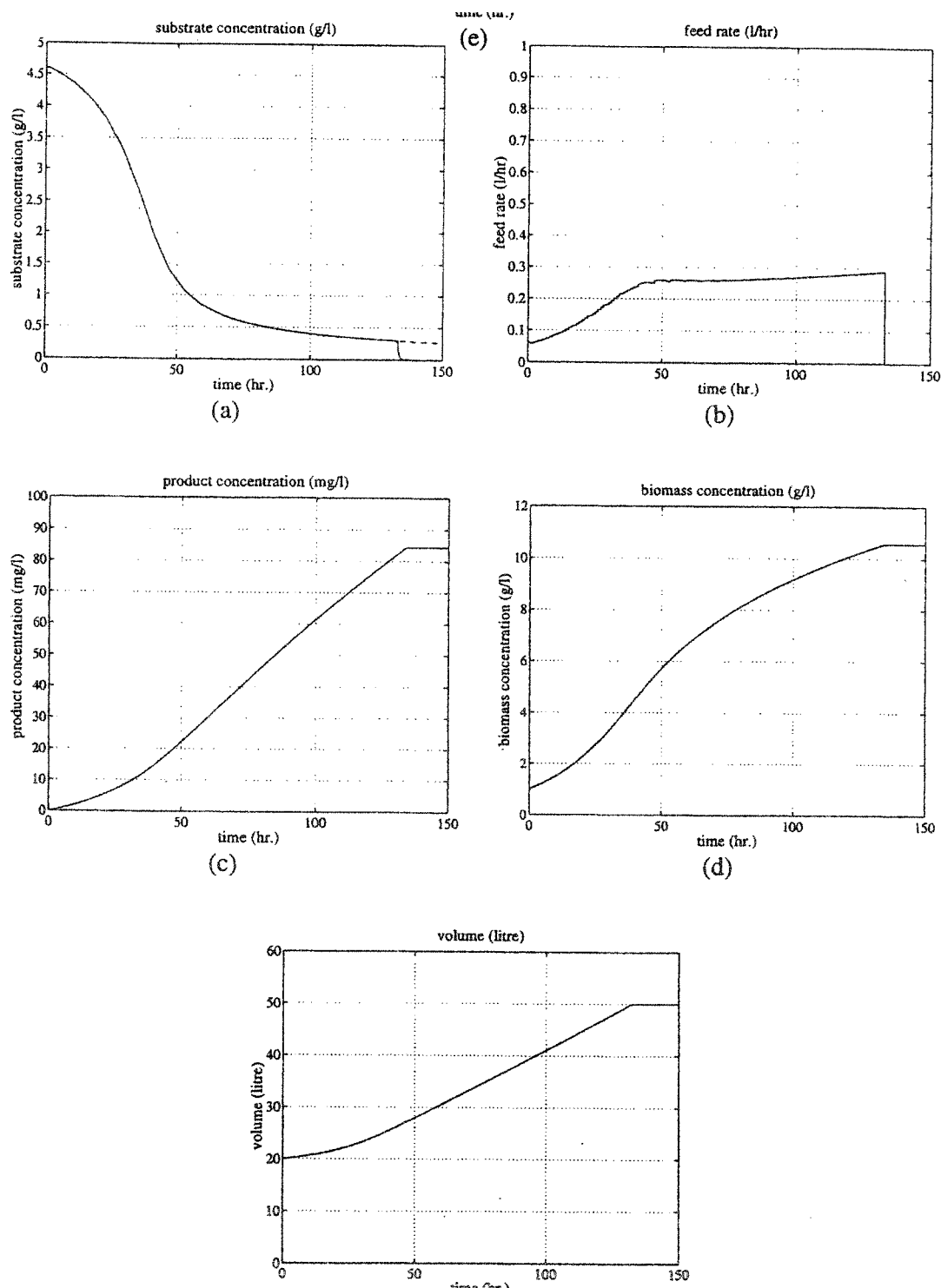
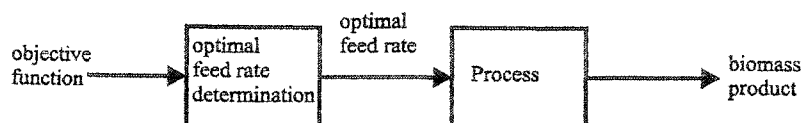


Fig 4 Simulation of closed-loop optimal control for a secondary metabolite process (assuming perfect model). Dashed line in (a) shows optimal substrate concentration profile. In the absence of plant-model mismatch or disturbance, both open-loop optimal feed rate and closed-loop optimal control algorithms lead to identical results

Fig 5 Block diagram for open-loop optimal feed rate profile method





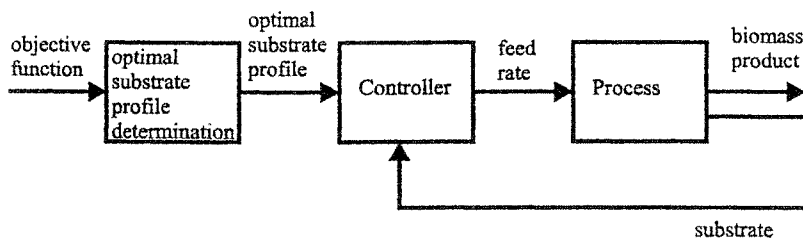


Fig 6 Block diagram for closed-loop optimal control method

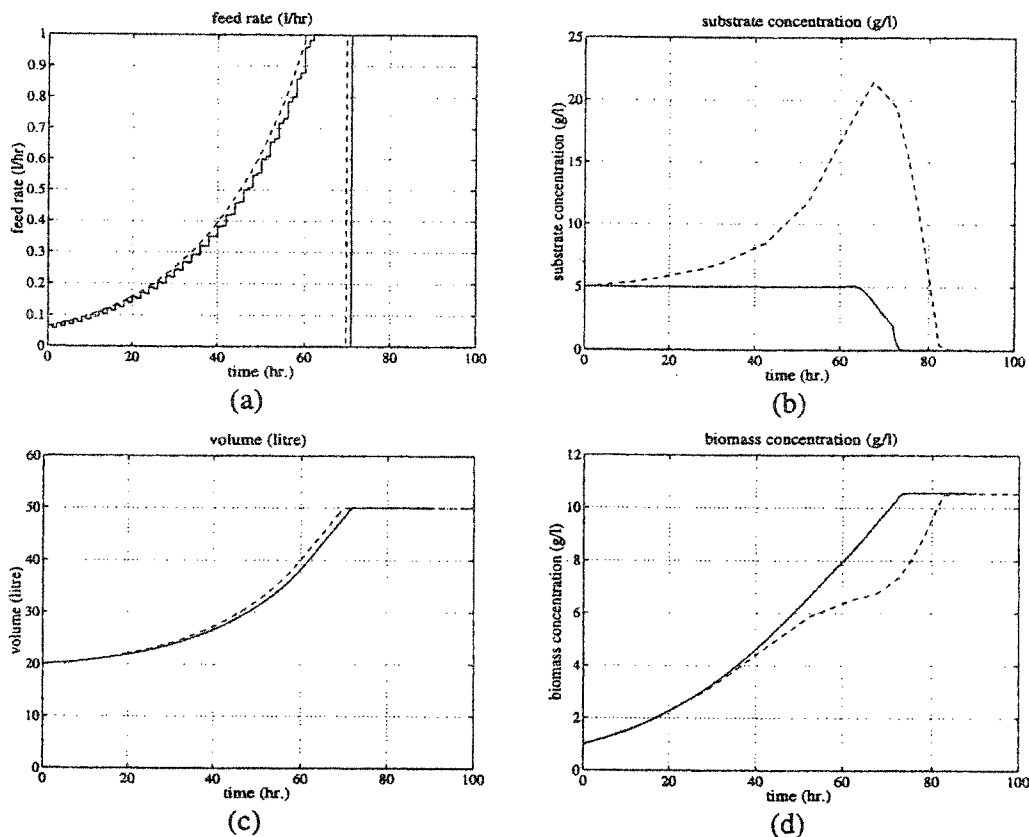


Fig 7 Comparison of both methods for plant-model mismatch in the primary metabolite production. Solid line = closed-loop optimal control method; dashed line = open-loop optimal feed rate profile method

the proposed method). This can be explained by the fact that in the open-loop method, the error in parameter  $Y_{xs}$  results in a slightly higher pre-calculated feed rate than the process needs. The miscalculated feed rate profile therefore diverges the substrate concentration in the fermenter from the optimal level at 5 g/l (Fig 7b). It should be noted that diverging the substrate concentration from the optimal level results in the lower specific growth rate. The micro-organisms at the lower specific growth rate consume a smaller amount of substrate and therefore need a lower substrate feed rate, while the pre-determined substrate feed rate still progressively increases. This in turn results in more divergence of the substrate concentration from the optimal level and less in the specific growth rate. This situation continues until the culture volume reaches the maximum and the feed rate stops. The substrate concentration then starts decreasing until depleted at the end of the batch (85 hr). This behaviour in the fermenter can be summarised as: the error in parameter  $Y_{xs}$  results in the higher feed rate, the higher feed rate results in the substrate concentration

diverging from the optimal level, diverging the substrate concentration from the optimal level reduces the specific growth rate, and low specific growth rate results in the process operating time being prolonged. More detail for the comparison of both methods for the primary metabolite production is in Vanichsriratana *et al* (1996).

For the secondary metabolite process, the open-loop method results in a lower final product concentration (80 mg/l) than the proposed closed-loop method (87 mg/l). This is also because the incorrect parameter causes the miscalculation of the optimal feed rate profile and therefore deviates the substrate concentration from the optimal production path, which results in the lower final product concentration.

For both primary and secondary metabolite processes, the error in parameter  $Y_{xs}$  has been compensated by feedback in the closed-loop optimal control method. For the primary metabolite production process, improvement of the closed-loop method over the open-loop optimal feed rate method at different levels of error in the parameter  $Y_{xs}$  is shown in Fig 9 as percent of shorter

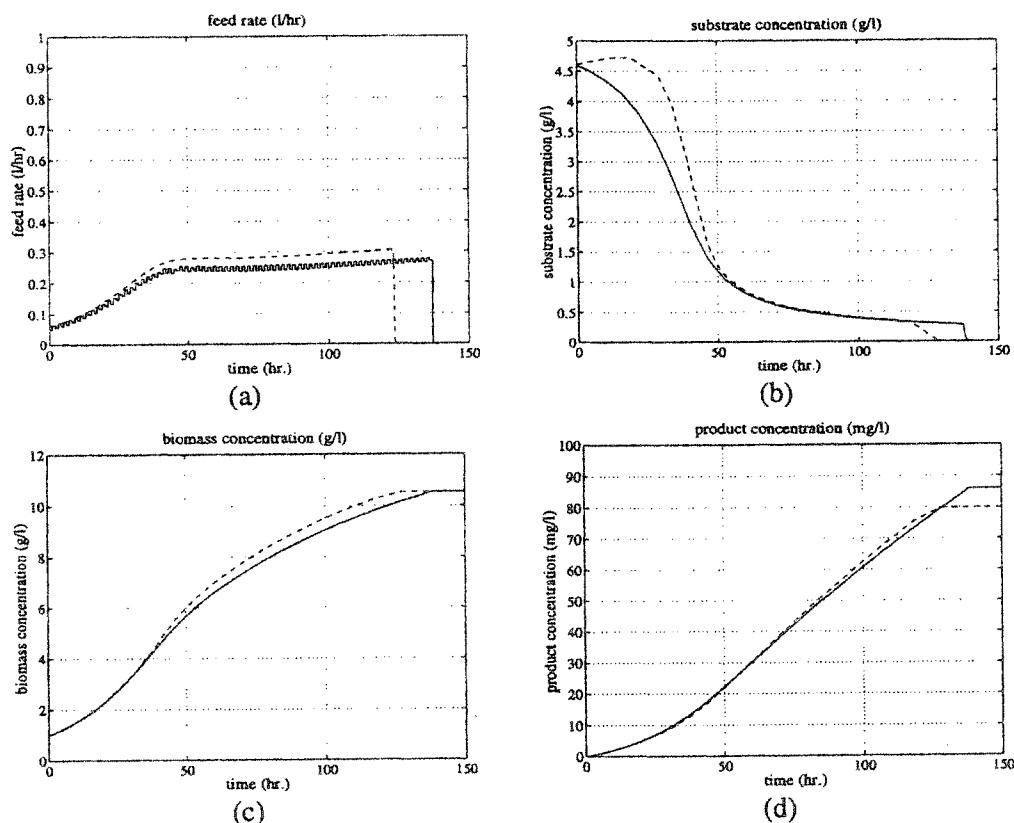


Fig 8 Comparison of both methods for plant-model mismatch in the secondary metabolite production. Solid line = closed-loop optimal control method; dashed line = open-loop optimal feed rate profile method

Fig 9 The closed-loop method results in a shorter operating time than the open-loop method in the primary metabolite production in the plant-model mismatch case

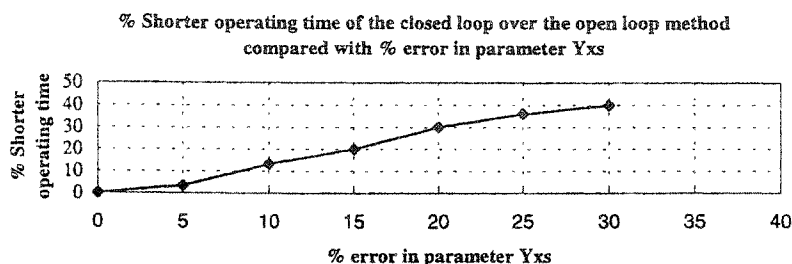
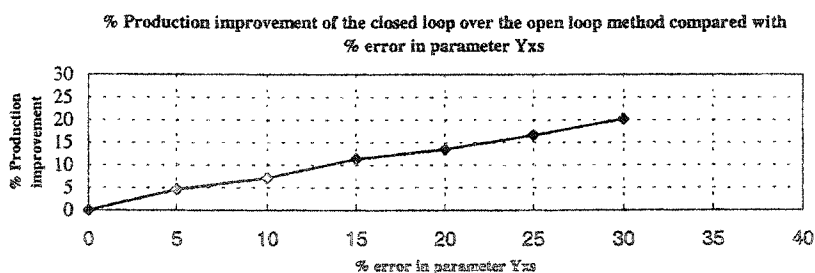


Fig 10 The closed-loop method results in production improvement over the open-loop method in the secondary metabolite production in the plant-model mismatch case



operating time. For the secondary metabolite production process, production improvement of the closed-loop over the open-loop optimal feed rate method at different levels of error in the parameter  $Y_{xs}$  is shown in Fig 10.

Moreover, the proposed closed-loop method provides the optimal substrate concentration profile which is specific to the process and is not affected by changes in feed rate constraints. Comparing to the open-loop optimal feed rate profile method, change in minimum or maximum feed rate would result in the different time

length in the combination of optimal feed rate sequence. This specific substrate concentration profile property thus provides more flexibility in its usage.

In the case of a time-varying process, given that the process model is known, the model predictive control can also be used to cope with this characteristic (Ogunnaike and Adewale, 1986). Also, as the process is time varying, the optimal substrate profile previously determined from the nonlinear model should be replaced by an on-line optimal substrate determination.

## 5.2 Effect of cost factor in the objective function

It is also interesting to note an effect of cost factor per unit of operating time ( $\varepsilon$ ) in the given objective function. For the biomass production shown in Section 4.1.1, the differences in cost factor have no effect in this process as the optimal substrate concentration profile is always constant at the level that maximises the specific growth rate. Note that maximising the specific growth rate maximises the biomass growth rate and, consequently, results in the shortest operating time in this process.

The situation is rather different for the secondary metabolite production in Section 4.1.2. The optimal substrate concentration profile (Fig 2) shows that the process should start with the biomass production phase (the substrate concentration at the highest is around 5 g/l, which is the optimal substrate concentration for the biomass production) and then progressively change to the product formation phase as the substrate concentration gradually decreases. As the cost factor increases, a longer time has been devoted for the biomass production (see also Fig 1), resulting in a shorter fermentation time. This shorter operating time, however, comes at the cost of a smaller amount of the secondary metabolite being produced. It is also worth noting that in the event of no cost factor ( $\varepsilon = 0$ ) in the objective function, the Hamiltonian equation changes to (for a free final time problem):

$$H = \lambda_X \mu X + \lambda_P \pi X = 0 \quad \dots (47)$$

and the optimal substrate concentration profile can be calculated from:

$$\frac{\partial H}{\partial S} = \lambda_X X \frac{\partial \mu}{\partial S} + \lambda_P X \frac{\partial \pi}{\partial S} = 0 \quad \dots (48)$$

Finding the value of  $\lambda_X$  in Eqn (47) and substituting into Eqn (48) yields:

$$\lambda_P (\mu \pi' - \pi \mu') = 0 \quad \dots (49)$$

As  $\lambda_P$  is not zero (see Eqns (33) and (35)), this implies that:

$$(\mu \pi' - \pi \mu') = 0 \quad \dots (50)$$

or

$$\frac{d(\pi/\mu)}{dS} = 0 \quad \dots (51)$$

Eqn (51) means that the optimal substrate concentration is to be kept at the level which maximises the ratio between the specific product formation rate and the specific growth rate. Note that operating time in this case can be very long, and despite a very high amount of secondary metabolite being produced, an excessively long operating time is not wanted in industry. Therefore, the cost factor ( $\varepsilon \neq 0$ ) should be incorporated into the objective function. More details on the effect of cost factor are in Vanichsriratana (1996).

## 6. Conclusion

The proposed method divides an optimisation problem in a fed-batch fermentation into two parts. First, the

optimal substrate concentration profile that optimises the given objective function is derived. Secondly, a controller is designed to track the obtained substrate concentration profile. The main advantage of the proposed method compared with the optimal feed rate profile method usually used in the literature is that the proposed method not only avoids a singular control problem but also transforms the problem to closed-loop control which performs more robustly in particular in the presence of model inaccuracies and unmeasurable disturbances.

Examples of primary and secondary metabolite production processes have been used to demonstrate and compare the performance of the proposed method with that of the open-loop optimal feed rate profile method. It is shown that both methods give very similar results when the process model is an exact representation of the plant, and that the proposed method gives much better results when the process model is not exactly matched with the plant.

The proposed method can be extended to a multi-input multi-output case, where the optimal profiles of temperature, pH, etc. can be derived and controlled besides the substrate concentration. This is an area requiring further investigation.

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## 8. References

- Ali, E., and Zafriou, E. 1993. 'Optimization-based tuning of nonlinear model predictive control with state estimation', *J Proc Cont*, **3**(2), 97–107.
- Banerjee, R. 1993. 'Evolutionary operation (EVOP) to optimize three-dimensional biological experiments', *Biotech Bioeng*, **41**(1), 67–71.
- Bastin, G., and Dochain, D. 1986. 'On-line estimation of microbial specific growth rates', *Automatica*, **22**(6), 705–709.
- Belgardt, K. H., Kuhlmann, W., Meyer, H. D., Schugerl, K., and Thoma, M. 1986. 'Application of an extended Kalman filter for state estimation of a yeast fermentation', *IEE Proc-D*, **133**(5), 226–234.
- Biegler, L. T. 1991. 'Optimization Approaches to Nonlinear Model Predictive Control'. In: *Chemical Process Control CPCIV*. Proceedings of the Fourth International Conference on Chemical Process Control. Padre Island, Texas: CACHE, AIChE.
- Cazzador, L. 1988. 'On the optimal control of fed-batch reactors with substrate-inhibited kinetics', *Biotech Bioeng*, **31**, 670–674.
- Cuthrell, J. E., and Biegler, L. T. 1987. 'On the optimization of differential-algebraic process systems', *AIChE J*, **33**(8), 1257–1270.
- Dunn, I. J., and Mor, J.-R. 1975. 'Variable-volume continuous cultivation', *Biotech Bioeng*, **17**, 1805–1822.
- Eaton, J. W., and Rawlings, J. B. 1990. 'Feedback control of chemical processes using on-line optimization techniques', *Comp Chem Eng*, **14**(4/5), 469–479.

- Green, M., and Limebeer, D. 1994. *Robust Control Systems*. Prentice Hall.
- Han, K., and Levenspiel, O. 1988. 'Extended monod kinetics for substrate, product, and cell inhibition', *Biotech Bioeng*, **32**, 430-437.
- Hong, J. 1986. 'Optimal substrate feeding policy for a fed batch fermentation with substrate and product inhibition kinetics', *Biotech Bioeng*, **28**, 1421-1431.
- Huber, F. M., and Tietz, A. J. 1983. 'Defined media strategies for the biosynthesis of Cephalosporin C', *Biotech Letter*, **5**(6), 385-390.
- Kennedy, M. J., Prapulla, S. G., and Thakur, M. S. 1992. 'Designing fermentation media: a comparison of neural networks to factorial design', *Biotech Tech*, **6**(4), 293-298.
- Lim, H. C., Chen, B. J., and Creagan, C. C. 1977. 'An analysis of extended and exponentially-fed-batch cultures', *Biotech Bioeng*, **19**, 425-433.
- Lim, H. C., Tayeb, Y. J., Modak, J. M., and Bonte, P. 1986. 'Computational algorithms for optimal feed rates for a class of fed-batch fermentation: numerical results for penicillin and cell mass production', *Biotech Bioeng*, **28**, 1408-1420.
- Liu, Y. C., Tsao, J. H., Chang, W. M., and Wu, W. T. 1992. 'On-line state estimation for L-Lysine fermentation'. In: *Modelling and Control of Biotechnical Processes*. Colorado, USA, 279-282.
- Modak, J. M., Lim, H. C., and Tayeb, Y. J. 1986. 'General characteristics of optimal feed rate profiles for various fed-batch fermentation processes', *Biotech Bioeng*, **28**, 1396-1407.
- Nahlik, J., and Burianec, Z. 1988. 'On-line parameter and state estimation of continuous cultivation by extended Kalman filter', *Appl Microbiol Biotechnol*, **28**, 128-134.
- Ohno, H., Nakanishi, E., and Takamatsu, T. 1976. 'Optimal control of a semibatch fermentation', *Biotech Bioeng*, **18**, 837-864.
- Ogunnaike, B. A., and Adewale, K. E. P. 1986. 'Dynamic matrix control for process systems with time varying parameters', *Chem Eng Commun*, **47**, 295-314.
- Park, S. H., Hong, K. T., Lee, J. H., and Bae, J. C. 1983. 'On-line estimation of cell growth for glutamic acid fermentation system', *Eur J Appl Microbiol Biotechnol*, **17**, 168-172.
- Parulekar, S. J., and Lim, H. C. 1985. 'Modeling, optimization and control of semi-batch bioreactors', *Adv Biochem Eng*, **32**, 207-258.
- Renfro, J. G., Morshedi, A. M., and Asbjornsen, O. A. 1987. 'Simultaneous optimization and solution of systems described by differential/algebraic equations', *Comp Chem Eng*, **11**(5), 503-517.
- Shimizu, H., Araki, K., Shioya, S., and Suga, K.-I. 1991. 'Optimal production of glutathione by controlling the specific growth rate of yeast in fed-batch culture', *Biotech Bioeng*, **38**, 196-205.
- Shimizu, H., Takamatsu, T., Shioya, S., and Suga, K. I. 1986. 'An algorithmic approach to construction of the on-line estimation system for the specific growth rate', *Biotech Bioeng*, **33**, 354-364.
- Sistu, P. B., and Bequette, B. W. 1991. 'Nonlinear predictive control of uncertain processes: application to a CSTR', *AIChE J*, **37**(11), 1711-1723.
- Stephanopoulos, G., and San, K. Y. 1982. 'On-Line Estimation of Time-Varying Parameters. Application to Biochemical Reactors'. In: *IFAC Modelling and Control of Biotechnical Processes*. Helsinki, Finland, 195-199.
- Takamatsu, T., Shioya, S., and Okada, Y. 1985. 'Profile control scheme in a bakers' yeast fed-batch culture', *Biotech Bioeng*, **27**, 1675-1686.
- Terwiesch, P., Agarwal, M., and Rippin, D. W. T. 1994. 'Batch unit optimization with imperfect modelling: a survey', *J Proc Cont*, **4**(4), 238-258.
- Vanichsriratana, W. 1996. 'Optimal control of fed-batch fermentation processes', PhD Thesis, University of Westminster, UK.
- Vanichsriratana, W., McFarlane, D., Keshavarz, T., and Leigh, J. R. 1996. 'Comparison of open loop optimal control and closed loop optimal control of a fermentation process', *International Control Conference* 96, UK, 258-263.
- Weigand, W. A., Lim, H. C., Creagan, C. C., and Mohler, R. D. 1979. 'Optimization of a repeated fed-batch reactor for maximum cell productivity', *Biotech Bioeng Symp*, **9**, 335-348.
- Yamane, T., and Shimizu, S. 1984. 'Fed-batch techniques in microbial processes', *Adv Biochem Eng*, **30**, 148-192.
- Zeng, A. P., Byun, T. G., and Deckwer, W. D. 1991. 'On-line estimation of viable biomass of a micro-aerobic culture using exit gas analysis', *Biotechnol Techniques*, **5**(4), 247-250.
- Zhang, B. S., Tang, R., Leigh, J. R., Dixon, K. D., and Hinge, R. D. 1996. 'Improved Quality and Productivity in Secondary Metabolite Fermentation through Estimation Control and Scheduling', *Proc 13th IFAC World Congress*. San Francisco.